PROVEIT / RESEARCH CONTRIBUTION

Explicit Diophantine formulations of Con(ZFC)

A fixed K215 specialization, a seven-equation system, and a polynomial in fifteen positive integer variables

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22 October 2026

Recommended readable form. Seven polynomial equations in sixteen positive integer variables, maximum degree 28, with a shared 64-operation evaluation circuit. Their sum of squared residuals is one polynomial of degree 56, evaluated in 84 operations.
Fewer variables. One polynomial in fifteen positive integer variables, degree 295, evaluated in 82 operations.
Meaning. Each formulation has a solution exactly when the fixed recursive proof system for ZFC has a finite refutation. Consequently its lack of solutions is equivalent, by mathematics formalizable in ZFC, to Con(ZFC).
Scope. This is an explicit mathematical construction with reproducible source and algebra audits. The inherited K215 machine endpoint has Lean and Rocq verification. The new Diophantine reduction and the inherited fixed-program polynomial compiler are not newly kernel checked by this submission. No claim of global minimality is made.

Source snapshot: `cdd582bec13f908df3711fa9e05d1ddac368a5db`.

All coefficients are fixed by finite formulas and the included literal machine table. Their enormous decimal expansions are not materialized.

Abstract

We specialize the ProveIt fixed-program Diophantine compiler to the exact 215-state K215 machine calibrated to ZFC refutation. Since only one input is needed, the universal construction's input Pell block can be replaced by equality to a fixed marker. We prove that the three deleted input witnesses can be restored as positive integers after main-rank recovery; the proof uses neither a decoded computation nor the power-recovery lemma. Combining this reduction with the repository's ordinary auxiliary projection gives seven equations in sixteen positive unknowns. Combining it with the normalized strong-root elimination gives one polynomial in fifteen positive unknowns. We specify an actual coefficient slice by a finite machine-to-window compiler, supply exact evaluation circuits, prove uniform degree bounds, and give unrestricted-integer and quartic variants. The distinction between a mathematical proof in ZFC, an external proof-assistant theorem, and a new end-to-end formalization is kept explicit throughout.

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1 Result, domain, and meaning of simplicity

Fix the recursively presented ZFC proof calculus used by the repository's K215 endpoint. Let $\mathrm{Ref}_{\mathrm{ZFC}}(p)$ mean that p codes a finite derivation of contradiction in this calculus, and put

$$\mathrm{Con(ZFC)} \ :\Longleftrightarrow \ \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

The source endpoint is the theorem `KEndpoint215.blank_halts_iff_zfc_refutation`; its refutation proposition is `BProvZFCax_s[]Form.fBot` [1, 2]. Coding finite syntax and finite derivations by natural numbers identifies this proposition with $\exists p\ \mathrm{Ref}_{\mathrm{ZFC}}(p)$.

Theorem 1.1 (Explicit fixed-instance result)**.** *With the fixed integers defined in Section 7, the system $\mathcal{S}_{16}$ in Section 3 and the polynomial P_{15} in Section 4 satisfy*

$$\mathrm{ZFC} \vdash \left(\exists \mathbf{z} \in \mathbb{Z}_{>0}^{16}\ \mathcal{S}_{16}(\mathbf{z})\right) \longleftrightarrow \exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p),$$
$$\mathrm{ZFC} \vdash \left(\exists \mathbf{z} \in \mathbb{Z}_{>0}^{15}\ P_{15}(\mathbf{z}) = 0\right) \longleftrightarrow \exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

Negating either equivalence gives the requested equivalence between lack of solutions and Con(ZFC)*. The same holds for the sum-of-squares and domain-conversion variants below.*

The theorem is an ordinary mathematical theorem, using the explicitly identified inherited compiler and Pell lemmas. Its proof does not use an assumption that ZFC is consistent. Section 10 explains the formalization boundary; the included executable checks are not a replacement for the mathematical proof.

Form	Positive unknowns	Equations	Degree	Operations
Recommended $\mathcal{S}_{16}$	16	7	28	64
Sum of squared residuals	16	1	56	84
Normalized product	16	1	157	74
Reduced auxiliary product	16	1	125	75
Ordinary auxiliary product	16	1	101	76
P_{15}, eliminated strong root	15	1	295	82
Eliminated-root comparison form	15	6	244	71

For a system, "degree" means the maximum degree of its residual polynomials after all abbreviations are expanded. "Operations" means binary additions, subtractions, and multiplications in the supplied shared evaluation circuit, including multiplication by a nontrivial constant. For a comparison system it counts computation of the comparison sides. It does not count expansion or construction of fixed coefficients. The circuits contain no division or variable exponentiation.

There is no single mathematical ordering called "simplest." Number of unknowns, degree, arithmetic circuit size, coefficient bit length, and proof complexity compete. This submission favors a short displayed system and a small fully specified circuit. It does not claim a minimum over all possible Diophantine representations. In particular, classical universal equations with nine nonnegative unknowns exist [8]; the present fifteen-variable result is not a variable-count record for general existence theorems.

2 Fixed constants and shared polynomial abbreviations

The fixed radix parameters are

$$b = 5^{42}, \qquad L_c = 5^{43}, \qquad d = bL_c = 5^{85}, \qquad B = 2^d, \qquad W_* = 2^b B^2 = 2^{2d+b}. \tag{1}$$

The other fixed integers K, M_C, M_F are the finite sums in Section 7. In particular,

$$K = D_C + BD_R, \qquad M_F = M_F^{\text{native}} + B - 1. \tag{2}$$

The shift by $B-1$ in M_F is essential. The source port called `MF` is not the native field mask.

The following definitions are shared by the main constructions:

$$q = (B-1)J + 1, \qquad X = wq, \qquad Y = sq^3, \qquad E = XY, \tag{3}$$

$$k = \eta + \zeta, \qquad c = kY + \eta, \qquad a = E + Y, \quad H = 4a + 3, \tag{4}$$

$$\Delta = a^2 + H = (a+2)^2 - 1, \qquad D_0 = X + ac + \gamma H, \tag{5}$$

$$C = q - F - Z - \alpha - 2d, \qquad W = C - Z, \tag{6}$$

$$R = (q^2 - Z - qF)(q^2 - 1) + (M_C + qM_F)J, \qquad L_0 = E(kY). \tag{7}$$

Define four useful factors

$$N_1 = \tau^2 - L_0(L_0 + k), \qquad N_2 = D_0^2 - \Delta c^2, \tag{8}$$

$$N_k = k - hE - R, \qquad N_t = (K + w)C + q - F - t(q-1). \tag{9}$$

Every letter on the left of these definitions is a polynomial abbreviation, not an additional unknown. Only the explicitly listed witnesses are quantified. Although those witnesses are positive, some computed abbreviations, such as C, W, R, can be negative away from a solution. Soundness proofs must establish the required bounds at a solution rather than assume them from notation.

3 The recommended seven-equation system

Take the sixteen strictly positive integer unknowns

$$(J, F, \alpha, t, f, h, i, T, s, w, \tau, \eta, \zeta, y, Z, \gamma) \in \mathbb{Z}_{>0}^{16}. \tag{10}$$

Besides the shared abbreviations, put

$$V = c(Tf - 1) - Rf^2, \qquad K_a = \Delta(f^2 - 1). \tag{11}$$

Our recommended system is

$$\boxed{\begin{aligned} \tau^2 - L_0(L_0 + k) &= 1, \\ D_0^2 - \Delta c^2 &= 1, \\ (ic^2)^2 &= K_a, \\ K_a(V^2 - y^2) + y^2 &= 1, \\ k - hE - R &= 1, \\ (K + w)C + q - F - t(q-1) &= 1, \\ W &= W_*. \end{aligned}} \tag{12}$$

This is literally a system over $\mathbb{Z}$ with sixteen positive unknowns; it contains no unbounded quantifier disguised as an abbreviation. The constants in its coefficients are fixed before any unknown is chosen. The shared circuit has 34 multiplications and 30 additions/subtractions. Its maximum residual degree is 28.

Let $G_1, \ldots, G_7$ be the left sides minus the right sides of (12), in displayed order. Then

$$\boxed{H_{16} = G_1^2 + G_2^2 + \cdots + G_7^2 = 0.} \tag{13}$$

Over integers this is equivalent to the seven equations. It has exact degree 56, and the supplied shared circuit uses 41 multiplications and 43 additions/subtractions, for 84 operations in total.

The witness i here belongs to the ordinary strong equation. The normalized construction used next has a different normalization of its witness also called i. We assert equality of the accepted proposition between these constructions, not coordinatewise equality of their solution sets.

4 A single polynomial in fifteen positive unknowns

Take the fifteen strictly positive unknowns

$$(J, F, \alpha, t, h, i, T, s, w, \tau, \eta, \zeta, y, Z, \gamma) \in \mathbb{Z}_{>0}^{15}.$$

Use (3)–(9), and define

$$N_W = 1 + (W - W_*)^2, \tag{14}$$

$$S = \Delta i c^2, \qquad Q = S^2, \qquad D_* = 1 + \Delta i^2 c^4, \tag{15}$$

$$b_* = c + RD_*, \qquad u_* = cT, \tag{16}$$

$$A_* = Q(D_* u_*^2 + b_*^2) - (Q-1)y^2 - 1, \qquad B_* = 2Q u_* b_*, \tag{17}$$

$$N = A_*^2 - D_* B_*^2. \tag{18}$$

The requested single polynomial can be taken to be

$$\boxed{P_{15} = N_1 N_2 N_W N_k N_t \,(N+1) - 1 = 0.} \tag{19}$$

Its degree is exactly 295. The supplied circuit evaluates it in 46 multiplications and 36 additions/subtractions. In particular, the apparently complicated local norm is computed using $D_* B_*^2 = 4(QD_* u_*^2)(Qb_*^2)$, sharing subexpressions already needed for A_*.

A six-equation alternative imposes

$$N_1 = N_2 = N_k = N_t = 1, \qquad W = W_*, \qquad A_*^2 = D_* B_*^2.$$

It has the same positive solvability, maximum degree 244, and a 71-operation shared comparison circuit. For exposition the seven-equation ordinary system is substantially easier to inspect.

5 Correctness: the fixed-marker reduction

5.1 The inherited interface, stated explicitly

The universal-input construction in the pinned repository represents acceptance of a fixed normalized machine using six fixed integer ports and positive witnesses. We use the complete75 compiler and its complete84/85/86/92 projections [3, 5, 6]. Its ordinary input is a positive integer x; for its input convention the unary word has length $2x + 2$. Our normalized machine ignores that word, so the choice $x = 1$ is sufficient.

In the complete84 source the main Pell root has the form

$$D_0 = X + ac + (\rho + \sigma)H,$$

and the input block enforces, at a zero,

$$W = 2^{2dx+b}.$$

It uses three positive witnesses δ, ρ, σ. The fixed-marker operation replaces $\rho+\sigma$ by a free positive γ, deletes the three witnesses, fixes $x = 1$, and imposes $W = W_*$; in a product polynomial the last equality is represented by $N_W = 1 + (W - W_*)^2$. Two quantified witnesses are saved overall.

The key inherited lemma is local to the Pell kernel. Write

$$(A + \sqrt{A^2 - 1})^n = \chi_A(n) + \psi_A(n)\sqrt{A^2 - 1}, \qquad A = a + 2.$$

Lemma 5.1 (Input-independent rank recovery)**.** *On the valid compiler slice of Section 7, the retained first and main Pell conditions, index and transport conditions, and ordinary auxiliary conditions of $\mathcal{S}_{16}$ give*

$$R \text{ odd}, \qquad D_0 = \chi_A(R), \qquad c = \psi_A(R), \qquad R > q. \tag{20}$$

The normalized auxiliary conditions

$$f^2 - \Delta i^2 c^4 = 1, \qquad QV^2 - (Q-1)y^2 = 1, \qquad Q = \Delta^2 i^2 c^4 \tag{21}$$

give the same conclusion when $f > 0$ and T is any integer. In the product versions, they also force $N_k = N_t = 1$ after the initial integer-unit argument.

This is the precise imported rank lemma, not an assertion that a finite test establishes all Pell solutions. The proofs are supplied as the pinned notes `review_complete85_auxiliary_bezout_math.md`, `complete84_signed_quotient_absorption.md`, and `review_complete86_ordinary_auxiliary_math.md`; the independent review in this archive traces which hypotheses each step uses.

For orientation, the raw packing and transport bounds established before rank recovery are

$$q \geq B \geq 16, \qquad 3q + 1 < R, \qquad R + 2 < q^4 \leq E < a. \tag{22}$$

The first equation can be written $\tau^2 - U(U+1)k^2 = 1$, where $U = XY^2$, and yields $k = 2\psi_{2U+1}(n)$. The main equation gives $c = \psi_A(p)$. Strong divisibility and the auxiliary congruences then give $R \equiv \pm p \pmod{c}$; the proved range bounds force $p = R$. The ratio $kY < c < k(Y+1)$ excludes the wrong index sign. These arguments use the old input factor only as an integer unit with value $+1$ and use $\rho + \sigma$ only through positivity. The marker factor is positive and the new γ is positive, so those hypotheses are preserved.

For the ordinary strong equation, the proof specifically establishes $p \mid m$ and $c \mid m$, where $f = \chi_A(m)$ and $ic^2 = \Delta\psi_A(m)$. It must not assume the stronger normalized divisibility $c^2 \mid \psi_A(m)$. The supplied ordinary proof uses only the weaker, valid relations. With seven separate equations the required factor values are imposed directly.

5.2 Restoration of the deleted positive witnesses

Lemma 5.2 (Fixed-marker restoration)**.** *Suppose the retained conditions give (20), and $W = W_*$. Then there exist positive integers δ, ρ, σ that restore the literal original input block at $x = 1$, with $\rho + \sigma = \gamma$.*

Proof. Put $u = 2d + b$. This is a fixed odd integer at least 3. The marker and the definitions of C, W give

$$q = F + 2Z + \alpha + 2d + W_* > W_* = 2^u > u.$$

Thus $R > u$, and since R, u are odd, $R \geq u + 2$. Define

$$\kappa = \psi_A(u), \quad \mu = \chi_A(u), \qquad \delta = \frac{\kappa - u}{\Delta}, \quad \rho = \frac{\mu - a\kappa - 2^u}{H}, \quad \sigma = \gamma - \rho. \tag{23}$$

These divisions occur only in the proof of witness existence, not in the polynomial.

Pell expansion modulo $\Delta = A^2 - 1$ gives $\psi_A(j) \equiv jA^{j-1} \pmod{\Delta}$. Since u is odd, $\kappa \equiv u \pmod{\Delta}$. Strict growth gives $\psi_A(u) > u$, so δ is a positive integer.

Let $E_A(j) = \chi_A(j) - a\psi_A(j)$. Its initial values are $1, 2$, and it satisfies

$$E_A(j+2) = 2AE_A(j+1) - E_A(j).$$

The sequence 2^j satisfies the same recurrence modulo $H = 4a + 3$, because $4 - 4A + 1 = -H$. Hence $E_A(j) \equiv 2^j \pmod{H}$. Also $E_A(2) = 4A - 1 > 4$ and $E_A(j+1) > 2E_A(j)$ for $j \geq 1$, so $E_A(u) > 2^u$. Therefore ρ is a positive integer.

It remains to prove $\sigma > 0$; this is the substantive restoration condition. From the main root, $E_A(R) = X + \gamma H$, while $E_A(u) = 2^u + \rho H$. Therefore

$$H(\gamma - \rho) = E_A(R) - E_A(u) - X + 2^u. \tag{24}$$

For $j \geq 1$, $E_A(j) = 2\psi_A(j) - \psi_A(j-1)$, so $\psi_A(j) < E_A(j) \leq 2\psi_A(j)$. Put $y_j = \psi_A(j)$. The recurrence and $y_{u-1} < y_u$ imply

$$y_{u+2} - 2y_u > (4A^2 - 2A - 3)y_u > A.$$

Since $R \geq u + 2$, $E_A(R) - E_A(u) > A > X$, the last inequality following from $a = Y(X+1) > X$. The right side of (24) is positive, proving $\sigma > 0$.

Finally,

$$\kappa = u + \delta\Delta, \quad \mu = W_* + a\kappa + \rho H, \quad \mu^2 - \Delta\kappa^2 = 1, \quad \gamma = \rho + \sigma.$$

These are exactly the deleted input relations. Every retained source row is unchanged, so the original positive input interface is restored. □

The converse is immediate from the original input theorem: at $x = 1$ the marker is W_*, and setting $\gamma = \rho + \sigma$ and deleting the three witnesses preserves every retained equation. Thus the fixed-marker specialization has exactly the same accepted proposition as the original fixed-input construction. This proves correctness of $\mathcal{S}_{16}$ through the ordinary projection theorem.

The restoration argument is deliberately ordered before power recovery, population counts, mask typing, or machine decoding. It therefore does not presuppose the conclusion that the modified equations already encode a computation.

6 Correctness: eliminating the strong root

Consider first the normalized sixteen-variable marker polynomial

$$P_{16}^{\mathrm{n}} = N_1 N_2 N_W N_k N_t \left[Q(V^2 - y^2) + y^2\right] \left[\Delta f^2 - Q\right] - \Delta, \qquad Q = \Delta^2 i^2 c^4. \tag{25}$$

Its 74-operation circuit is the fixed-marker reduction of complete84. At a zero one first divides the equality by the positive common factor Δ, then applies the integer-unit and sign arguments of the normalized kernel. It would be incorrect to infer that factors whose product is Δ are individually units without that normalization.

The complete92 norm trick can now eliminate f. We include the full local algebra because its sign issue matters.

Lemma 6.1 (Integral reconstruction)**.** *Every positive zero of P_{15} reconstructs a unique nonzero integer $f = A_*/B_*$, satisfying $f^2 = D_*$ and the normalized auxiliary equations. This integer is in fact positive.*

Proof. First, $D_* > c > 0$. If $R \geq 0$, then $b_* = c + RD_* > 0$; if $R \leq -1$, then $b_* \leq c - D_* < 0$. Hence $B_* = 2QcTb_*$ never vanishes, even before the equations are imposed.

At a zero of (19), both $N_1N_2N_WN_kN_t$ and $N+1$ are integer units. Since B_* is even,

$$N + 1 \equiv A_*^2 + 1 \pmod 4.$$

The residue is 1 or 2, and cannot be -1. Thus $N = 0$ and the other product is 1. The rational number $f = A_*/B_*$ satisfies $f^2 = D_* \in \mathbb{Z}$ and is therefore an integer: reducing the fraction to lowest terms shows its denominator must be 1.

From $f^2 = D_*$ one gets $\Delta f^2 - Q = \Delta$. Moreover $V = u_*f - b_*$, and

$$\begin{aligned} QV^2 - (Q-1)y^2 - 1 &= Q(D_*u_*^2 + b_*^2) - (Q-1)y^2 - 1 - 2Qu_*b_*f \\ &= A_* - B_*f = 0. \end{aligned}$$

Thus the reconstructed signed f gives a zero of (25).

Suppose $f < 0$. The simultaneous change $(f, T) \mapsto (-f, -T)$ preserves all expressions in the reconstructed normalized sixteen-variable source, because f, T occur there through f^2 and Tf. Now $f > 0$ and $T < 0$. Apply the signed version of Lemma 5.1, and then Lemma 5.2. This produces a zero of the literal original complete84 source with all its witnesses positive except possibly T. The repository's source-specific signed quotient theorem proves that every such original zero on a valid compiler slice has $T > 0$ [7]. This is a contradiction. The signed quotient theorem is applied only after the original input interface has been restored, so no positive-domain theorem is applied prematurely. □

Conversely, a positive zero of the normalized sixteen-variable source satisfies $f^2 = D_*$ and $A_* = B_*f$. Deleting f gives $N = 0$ and (19). Therefore P_{15} and the normalized marker source have the same positive solvability. Together with Lemma 5.2 and the fixed-program compiler this proves the fifteen-variable part of Theorem 1.1.

The signed-quotient lemma is a genuine mathematical dependency. An algebraic square-root reconstruction alone does not prove that its reconstructed root is positive. Its pinned proof and independent review are included in `vendor/`.

7 A fully determined coefficient slice

This section removes the common ambiguity "choose coefficients representing the desired recursively enumerable set." Every coefficient here is fixed by a literal finite table and finite arithmetic. The algorithm does not search for a halting computation or a ZFC refutation. The original table has SHA-256

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

It is included as `source/k215.tm`.

7.1 From the literal K215 table to a normalized machine

K215 has 215 nonhalting states, two symbols, blank symbol 0, initial state 0, and external halt state 215. Its 430 instructions are preserved exactly. We construct a machine with 436 states and six symbols. Its ordered states are

$$\text{start}, \text{enter}, \text{erase}, \text{return}, \text{dead}, \text{halt}, p_0, \ldots, p_{214}, m_0, \ldots, m_{214}.$$

The ordered alphabet is $0, 1, 2, 3, 4, 5$. At physical position $-n$, $n > 0$, symbols $0, 1, 4, 5$ respectively store the two-track pairs $(0, 0), (0, 1), (1, 0), (1, 1)$ for K215 positions $(n, -n)$. At the origin, symbols $2, 3$ store the origin bit $0, 1$.

State/read	New state	Write	Move
start/1	enter	2	stay
enter/2	erase	2	left
erase/1	erase	0	left
erase/0	return	0	right
return/0	return	0	right
return/2	p_0	2	stay

Unspecified prefix transitions go to dead, preserving the symbol and staying. Dead and halt are absorbing. A state p_q or m_q simulates K215 instruction q on the selected track. Positive-track movement is reversed physically, negative-track movement is unchanged. At an origin marker, either side state reads and updates the origin bit, moves left, and chooses the positive or negative side from K215's direction. A K215 target state 215 becomes halt. All cases are literal in `artifacts/folded_machine.json`.

On a positive unary word of length ℓ ending at the origin, initialization takes exactly $2\ell + 2$ steps and leaves a blank simulated tape, state p_0, and head at the origin. The invariant maps a K215 head position h to physical position $-|h|$ and the selected side to the sign of h; each simulated transition preserves it. Induction therefore proves halting equivalence. For the chosen ordinary input $x = 1$, the compiler's unary length is 4. The normalized machine accepts that input exactly when K215 halts.

7.2 The exact finite set of windows

The compiler alphabet has $a_t = 2628$ tiles, ordered as follows: two boundary tiles; all pairs of a tape symbol with a head label in {None, the 436 states}, in lexicographic order; then the four initial-row phase tiles L, I, Q, R with payloads $(0, \text{None}), (1, \text{None}), (1, \text{start}), (0, \text{None})$. The radius-one predicate is the exact function `local_valid` in `source/compiler_k215.py`. It is the pinned helical tableau predicate with stationary first-step markers [4].

The predicate preserves vertical boundaries, synchronizes horizontal boundaries, checks first-row phases, checks the three-neighbor machine update with at most one arriving head, forbids crossing a vertical boundary, and requires a last-row head to be halted. Its complete boundary cases are supplied as executable finite logic, with no call to a machine-acceptance oracle. This finite function is part of the coefficient definition.

Enumerate valid crosses (c, l, r, d, u) lexicographically. For each, enumerate the four corners (nw, ne, sw, se) lexicographically, yielding row-major windows $(nw, d, ne, l, c, r, sw, u, se)$. Move these two valid windows to the front, in the displayed order, omitting their later occurrences:

$$w_0 = (1, 1, 1, 2625, 2625, 2626, 439, 439, 878),$$
$$w_1 = (1, 1, 1, 2624, 2625, 2625, 2, 439, 439).$$

There are exactly

$$n_\times = 43\,126\,116\,188\,949 \quad \text{valid crosses},$$
$$n_w = n_\times a_t^4 = 2\,057\,035\,525\,715\,746\,135\,339\,455\,744 \quad \text{windows}.$$

The count is obtained by factoring the finite sum over left and right neighbors according to which head, if any, they send into the center. For example, when the center sends no stationary head, the interior count is

$$L_0 R_0 U_{s,0} + R_0 \sum_q L_q U_{s,q} + L_0 \sum_q R_q U_{s,q}.$$

The script treats initialization and boundary cases separately. It records each center's count. The factorization proof and the exact conventions are documented in `proofs/COMPILER.md`.

7.3 Layout and coefficient formulas

Set $A_c = 2^{92}$. This satisfies $A_c > n_w + 1$ and $A_c \equiv 1 \pmod 5$. Let

$$\begin{aligned} h_0 &= 9a_t + 4, & m_0 &= 184 + 30a_t + 8, & & \\ p_c &= \left\lceil \frac{n_w + 9a_t + 4 - m_0}{2} \right\rceil, & h_c &= h_0 + p_c, & m &= m_0 + 2p_c, \\ e_* &= n_w + 12a_t, & M &= m + 6a_t - 3. & & \end{aligned}$$

There is exactly one ignored dummy position. Important numerical values are

h_0	23 656
p_c	1 028 517 762 857 873 067 669 700 184
h_c	1 028 517 762 857 873 067 669 723 840
m	2 057 035 525 715 746 135 339 479 400
e_*	2 057 035 525 715 746 135 339 487 280
M	2 057 035 525 715 746 135 339 495 165

Write

$$\mathsf{H} = 51M + 3a_t + 1, \quad \mathsf{T}_1 = 105M + 4a_t + 2, \quad \mathsf{T}_2 = 159M + 4a_t + 3, \quad g = 186M + 4a_t + 4.$$

Use the fixed b, L_c, d, B from (1), and put $r_c = 2^b$. Define the finite set

$$\mathcal{E} = \{0, \dots, n_w - 1\} \cup \{n_w + 3a_t, \dots, n_w + 12a_t - 1\} \cup \{e_*\} \cup \{M, 3M, 9M, 27M\}.$$

For $0 \le r, c < 3$ and $0 \le s < a_t$ let

$$j(r,c,s) = 1 + (3r + c)a_t + s, \qquad e(r,c,s) = n_w + 3a_t + (2 - r)3a_t + (2 - c)a_t + s.$$

The coefficient attached to each exponent in $\mathcal{E}$ is

$$\begin{aligned} c_i &= 1 + \sum_{r,c=0}^{2} A_c^{j(r,c,w_i[3r+c])} + \sum_{t=1}^{4} A_c^{9a_t+t} && (0 \le i < n_w), \\ c_{e(r,c,s)} &= A_c^{j(r,c,s)}, && c_{3^t M} = A_c^{9a_t+1+t} \quad (0 \le t < 4), \\ c_{e_*} &= 0, && \mu_c = A_c - 2 + \sum_{j=1}^{h_c} A_c^j. \end{aligned}$$

All exponent ranges in these sums are fixed finite ranges, not Diophantine variables. Finally define

$$D_C = r_c^{3a_t} + r_c^{\mathsf{H}+a_t} + r_c^{8M} + r_c^{24M} + \sum_{e\in\mathcal{E}} c_e(r_c^{\mathsf{T}_1-e} + r_c^{\mathsf{T}_2-e}), \tag{26}$$

$$D_R = r_c^{\mathsf{H}}, \tag{27}$$

$$M_C = B - 1 - \sum_{e\in\mathcal{E}\setminus\{1\}} r_c^e - 2r_c^{e_*}, \tag{28}$$

$$\begin{aligned} M_F^{\text{native}} &= \mu_c(r_c^{\mathsf{T}_1} + r_c^{\mathsf{T}_2}) + \sum_{r=0}^{1}\sum_{c=0}^{2}\sum_{s=0}^{a_t-1} r_c^{e(r,c,s)} \\ &\quad + \sum_{r=0}^{2}\sum_{c=0}^{1}\sum_{s=0}^{a_t-1} r_c^{\mathsf{H}+e(r,c,s)} + r_c^{9M} + r_c^{27M} + 4. \end{aligned} \tag{29}$$

Together with (2), these specify every coefficient port of the displayed equations.

7.4 Why this is an admissible compiler instance

The complete75 compiler allows a power-of-two clause radix greater than n_w+1, extra zero clauses, and a sufficiently large power-of-five inner radix exponent. The above choices obey these rules. In particular, $L_c > g + 27M$, so all duplicated bands and anchors are separated. The sum of center coefficients is bounded by

$$\sum_e c_e \le (14n_w + 9a_t + 4)A_c^{h_0}.$$

The stronger modified75 no-carry margins follow by choosing b larger than binary exponent bounds for both $2\mu_c + 4$ and

$$4(m+3)\big(2\sum_e c_e + 6\big) + 8.$$

The two bounds computed in the certificate are 94 623 634 182 924 322 225 614 593 282 and 2 176 541, respectively; $b = 5^{42}$ exceeds both. Since $A_c \equiv 1$ and $r_c \equiv 2 \pmod 5$, direct evaluation of the finite coefficient recipe gives $2D_C - D_R \equiv 2 \pmod 5$. Thus the optional high-monomial correction is unnecessary. Both b and L_c are powers of five, as required for the inherited actual-index dummy-control argument.

The modifications $-2r_c^{e_*}$ and $+4$ in (28)–(29) belong to the complete75 compiler and must not be replaced by the older complete76 masks. The padded clause radix 2^{92} and conservative $b = 5^{42}$ increase coefficient size; they do not change the local predicate or accepted computation.

Thus the inherited fixed-program theorem applies to this actual machine and these actual fixed constants. The complete chain is

$$\begin{aligned} \text{positive Diophantine solution} &\longleftrightarrow \text{normalized machine accepts } 1^4 \\ &\longleftrightarrow \text{K215 halts} \\ &\longleftrightarrow \exists p\ \mathrm{Ref}_{\mathrm{ZFC}}(p). \end{aligned}$$

The first equivalence uses the proved marker restoration; the second uses the explicit fold-and-erase simulation; the third uses the K215 calibration.

8 Exact degree and operation certificates

Formal maximum-degree propagation through a circuit can overestimate degrees after cancellation. Here the uniform degrees are proved algebraically, and finite-field univariate evaluations serve only as additional diagnostics.

Give every witness degree 1 and every fixed numeral degree 0. Set

$$Q_0 = (B-1)J, \quad k_0 = \eta + \zeta, \quad H_0 = Q_0 - F - 2Z - \alpha, \quad T_2 = w(Q_0 - F - Z - \alpha) - tQ_0.$$

The leading homogeneous forms of N_1, N_2, N_k, N_t, N_W are respectively

$$-k_0^2 w^2 s^4 Q_0^{14}, \quad 8\gamma k_0 w^2 s^3 Q_0^{11}, \quad -hwsQ_0^4, \quad T_2, \quad H_0^2.$$

Their degrees are $22, 18, 7, 2, 2$. For the main norm the degree-22 terms cancel exactly; writing $g_0 = \gamma H$ gives the identity

$$(X + ac + g_0)^2 - (a^2 + H)c^2 = X^2 + 2Xg_0 + g_0^2 + 2acX + 2acg_0 - Hc^2,$$

which exposes its degree-18 leading term.

In $\mathcal{S}_{16}$, the auxiliary residual has degree 28 and leading form

$$k_0^2 w^2 s^4 Q_0^{14} T^2 f^4.$$

The other residuals have degrees $22, 18, 22, 7, 2, 1$, in the corresponding displayed order. Hence the sum of squares has exact degree 56 and leading form

$$k_0^4 w^4 s^8 Q_0^{28} T^4 f^8.$$

The coefficient of $\eta^4 w^4 s^8 J^{28} T^4 f^8$ is $(B-1)^{28} \neq 0$.

For P_{15}, Q, D_*, b_*, u_* have degrees $46, 34, 38, 6$. A_* has degree 122, while B_* has degree 90. Therefore $N = A_*^2 - D_* B_*^2$ has degree 244: the second term has only degree 214. The outer product has degree 51. The leading form of P_{15} is

$$8h\gamma i^{12} k_0^{27} w^{21} s^{48} Q_0^{177} (Q_0 - F)^4 H_0^2 T_2. \tag{30}$$

In particular the coefficient of

$$J^{184} h\gamma i^{12} \eta^{27} w^{21} s^{48} t$$

is $-8(B-1)^{184}$, uniformly nonzero. Its degree is 295.

The JSON sources list every binary gate, and the independent checker reconstructs every source row from the pinned parent packets. The full 82-gate and 64-gate schedules are reproduced in the appendices. Additional product variants have degrees $157, 125, 101$ and operation counts $74, 75, 76$ respectively; their full circuits and leading-form proof are included in `artifacts/` and `proofs/`.

9 Nonnegative and unrestricted integer versions

9.1 Nonnegative variables without changing their number

If $\mathbb{N} = \{0, 1, 2, \ldots\}$ is the preferred Diophantine domain, replace every positive witness z by $n + 1$, with $n \in \mathbb{N}$. This preserves the number of variables and the total degree, and preserves solvability in both directions. In particular it gives a fifteen-nonnegative-variable polynomial equivalent to refutability.

9.2 Three unrestricted integers for each positive witness

Define

$$\mathrm{Pos}(a, b, c) = 1 + a^2 + b^2 + c(c+1). \tag{31}$$

Its image on $\mathbb{Z}^3$ is exactly $\mathbb{Z}_{>0}$. Positivity is immediate since $c(c+1) \geq 0$ for every integer c. Conversely, for $z > 0$, Legendre's three-square theorem applies to $4z-3$, a positive integer congruent to 1 or 5 modulo 8. Modulo 4, a sum of three squares equal to 1 has exactly one odd square. Writing the two even roots as $2a, 2b$ and the odd root as $2c+1$ yields $z = 1+a^2+b^2+c(c+1)$ [10].

Substitution of independent copies of (31) gives

Form over $\mathbb{Z}$	Integer unknowns	Equations	Degree
$P_{15}(\mathrm{Pos}, \ldots, \mathrm{Pos})$	45	1	590
$\mathcal{S}_{16}(\mathrm{Pos}, \ldots, \mathrm{Pos})$	48	7	56
$H_{16}(\mathrm{Pos}, \ldots, \mathrm{Pos})$	48	1	112

The degree doubles exactly: restricting each triple to $(a, 0, 0)$ gives the substitution $z \mapsto 1 + a^2$, which sends a nonzero leading form to a nonzero form of twice its degree.

9.3 A completely explicit quartic alternative

Introduce one signed integer register for each of the 64 gates of the ordinary comparison circuit and keep the 16 original inputs. For each input z add three signed integers and the quadratic equation $z = \mathrm{Pos}(a, b, c)$. Each gate equation is linear or quadratic; add the seven output comparisons. This gives

$$16 + 64 + 3 \cdot 16 = 128 \quad \text{integer unknowns,} \qquad 64 + 7 + 16 = 87 \quad \text{quadratic equations.}$$

Their sum of squared residuals is a single quartic polynomial. The degree is exactly four, since a fresh positivity coordinate's fourth power occurs with coefficient 1. Solvability is unchanged: evaluating the gates extends a solution uniquely on registers, and the gate equations force the intended evaluation in the converse direction. This is an explicit degree-oriented alternative, not a claim to improve classical existential quartic bounds. Complete equation trees and the conversion generator are included in `artifacts/domain_variants.json` and `source/domain_variants.py`.

10 What is proved, checked, and still to formalize

Three different kinds of evidence must be distinguished.

Mathematical proof. The new fixed-marker restoration and strong-root argument are mathematical proofs. The historical compiler, Pell rank, and signed-quotient proofs are identified precisely and included as pinned source notes. The fold-and-erase machine reduction is proved by a finite transition invariant. The Diophantine and simulation parts use finite syntax, natural-number induction, elementary number theory, and finite computation coding. The K215 calibration additionally uses set-model arguments, described below. All of these mathematical arguments are available in ZFC.

Inherited formal endpoint. The repository supplies Lean and Rocq theorems calibrating K215 halting to ZFC refutation. Its README reports complete builds and states the proof-assistant axiom reports. Those are inherited results, not builds rerun for this submission. An external Lean or Rocq theorem is not by itself an encoded ZFC derivation. Nor does the absence of user axioms establish a general conservativity theorem for the assistant's foundations. The internal ZFC claim concerns the particular syntactic, simulation, and set-model calibration arguments, which can be carried out in ZFC; it does not invoke a global soundness theorem for Lean or Rocq.

The actual calibration first compares the finite `C222` calculus with ZF using first-order completeness, a seed model, a bisimulation quotient, and a well-founded submodel. It then compares ZF and ZFC refutations using the constructible submodel of a supplied set model of ZF, together with first-order soundness and completeness. These are ordinary ZFC arguments about set-sized domains, their subsets, quotients, and recursively defined satisfaction relations. They do not require a model of the entire proof-assistant universe hierarchy or an inaccessible cardinal. In particular, the argument never asserts unconditionally that ZFC has a model. The exact intermediate files and the set-domain interpretation are described in `proofs/FORMAL_STATUS.md`.

The source presentation matches standard ZFC: it has the usual ZF axiom schemas and Zermelo's selector form of Choice. Partial-functional Replacement is equivalent to the usual total-functional presentation by adding a default value and then separating the genuine range. The selector form of Choice is equivalent to the choice-function form by using disjoint tagged copies of a set-indexed family. Finite schema annotations make numerical proof checking primitive recursive. Thus the repository's syntactic consistency statement is provably equivalent in ZFC to that for a standard effective first-order presentation.

New executable audits. The supplied standard-library Python checks reconstruct the exact literal source transformations, verify dependency closure and operation counts, verify source hashes and coefficient layout inequalities, and perform independent algebra and degree diagnostics. The finite simulation checks cover 5160 local folded transitions and seven initialization lengths $1, 2, 3, 4, 7, 19, 64$. Both marker windows pass the local predicate. A separate exhaustive 18-tile fixture checks all 1 889 568 crosses and finds 121 165 valid crosses, agreeing with the factored count at every center. This fixture corroborates the count implementation; it is not a proof by testing of the unbounded compiler theorem.

No new Lean or Rocq kernel proof of the complete Diophantine theorem is delivered here. In particular, the repository's mathematical compiler notes remain a proof dependency. A later formalization should encode the new marker lemma and ordinary/normalized interfaces, then connect them to a formally verified machine-to-polynomial compiler. The exact coefficient recipe and finite source circuits supplied here make that task concrete and version-stable.

The statement is also meaningful in nonstandard models of ZFC. Internally quantified proof codes and finite computations need not be standard externally; the proof must relate them by the same arithmetic coding. It is not enough to argue only about standard computations. The use of the complete ZFC proof, including uniform induction, syntactic simulation, and set-model calibration, supports the internal equivalence without an appeal to the truth of consistency.

If ZFC is consistent, the displayed equations have no standard solutions. By Gödel's second incompleteness theorem, under the usual recursive-presentation hypotheses ZFC then cannot prove their lack of solutions. Mere consistency does not by itself justify the stronger claim that ZFC also cannot prove that a solution exists; a suitable soundness assumption would address that different assertion.

11 Reproduction, integration, and comparison

The archive is self-contained for its arithmetic and source checks and uses only Python's standard library. From its root run

```
python3 source/verify.py
python3 source/render_circuits.py
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

The first command regenerates the compiler and circuit outputs in a temporary directory, checks them against the delivered files, verifies the pinned vendor hashes, reruns the independent source

audit, and checks the domain variants. It does not expand the giant fixed integers, enumerate all n_w windows, search for a ZFC contradiction, or rerun the inherited assistant builds.

Path	Contents
`report.tex`, `report.pdf`	This report and its editable source.
`source/`	Literal K215 table, coefficient compiler, circuit transformation, independent audit, domain conversion, and reproduction driver.
`artifacts/`	Complete machine table, finite coefficient recipe, all polynomial circuits and comparison systems, integer-domain variants, and generated LaTeX gate tables.
`proofs/`	Detailed new proof, independent mathematical review, coefficient proof, domain proof, and provenance notes.
`vendor/`	Pinned upstream proof notes, original circuit packets, endpoint sources, manifest, and inherited notices.
`checks/`	Reproducible validation receipts and delivery manifest.

A natural repository destination is a new research subdirectory under `Computability/HilbertTenthProblem/Papers/research-wip/`, with the existing native-stream-queue files cited by the supplied manifest. No existing repository files need to be overwritten. The archive's own copies of upstream notes are frozen for review; their relative cross-links may refer to the original repository layout.

The universal complete84 and complete92 sources use respectively 18 and 17 positive witnesses. The new fixed-marker reduction removes two witnesses in each case and reduces each circuit by ten operations. The ordinary seven-equation form also removes unnecessary product finalization and compares strong-equation sides directly. The savings arise from the fixed input, so they must not be reported as new bounds for a varying universal input.

Classical work supplies small universal variable/degree pairs [8], and dedicated work represents provability Diophantinely [9]. This contribution's distinctive content is its exact K215 specialization, concrete finite coefficient recipe, proved removal of the input block, short shared circuits, and explicit review boundary. Its greatest remaining cost is coefficient size: $B = 2^{5^{85}}$ alone has $5^{85} + 1$ binary digits. A reduction in machine or tableau encoding size could improve this cost without changing the number of Diophantine unknowns.

A Literal circuit for the fifteen-variable polynomial

Each r_j in this appendix is a polynomial abbreviation. It is not an extra quantified variable. The table is generated directly from the delivered `fixed_marker82` JSON packet. All six named coefficient ports have the fixed values specified in Section 7.

Table 1: The exact 82-operation circuit for P_{15}.

Gate	Polynomial abbreviation	Gate	Polynomial abbreviation
1	$r_1 = \tau \cdot \tau$	42	$r_{42} = r_2 \cdot r_{30}$
2	$r_2 = (B-1) \cdot J$	43	$r_{43} = r_{42} + r_{31}$
3	$r_3 = r_2 + 1$	44	$r_{44} = r_4 - 1$
4	$r_4 = r_3 \cdot r_3$	45	$r_{45} = r_{43} \cdot r_{44}$
5	$r_5 = r_4 \cdot r_3$	46	$r_{46} = r_3 \cdot M_F$
6	$r_6 = w \cdot r_3$	47	$r_{47} = M_C + r_{46}$
7	$r_7 = s \cdot r_5$	48	$r_{48} = r_{47} \cdot J$
8	$r_8 = r_6 \cdot r_7$	49	$r_{49} = r_{45} + r_{48}$
9	$r_9 = \eta + \zeta$	50	$r_{50} = r_{41} - r_{49}$
10	$r_{10} = r_9 \cdot r_7$	51	$r_{51} = K + w$
11	$r_{11} = r_8 \cdot r_{10}$	52	$r_{52} = r_{51} \cdot r_{33}$
12	$r_{12} = r_{11} + r_9$	53	$r_{53} = r_{52} + r_{30}$
13	$r_{13} = r_{11} \cdot r_{12}$	54	$r_{54} = t \cdot r_2$
14	$r_{14} = r_1 - r_{13}$	55	$r_{55} = r_{53} - r_{54}$
15	$r_{15} = r_{10} + \eta$	56	$r_{56} = i \cdot r_{27}$
16	$r_{16} = r_8 + r_7$	57	$r_{57} = r_{56} \cdot r_{56}$
17	$r_{17} = r_{15} \cdot r_{16}$	58	$r_{58} = i \cdot r_{26}$
18	$r_{18} = r_6 + r_{17}$	59	$r_{59} = r_{58} \cdot r_{56}$
19	$r_{19} = 4 \cdot r_{16}$	60	$r_{60} = r_{59} + 1$
20	$r_{20} = r_{19} + 3$	61	$r_{61} = r_{49} \cdot r_{60}$
21	$r_{21} = \gamma \cdot r_{20}$	62	$r_{62} = r_{15} + r_{61}$
22	$r_{22} = r_{18} + r_{21}$	63	$r_{63} = r_{15} \cdot T$
23	$r_{23} = r_{22} \cdot r_{22}$	64	$r_{64} = r_{63} \cdot r_{63}$
24	$r_{24} = r_{16} \cdot r_{16}$	65	$r_{65} = r_{60} \cdot r_{64}$
25	$r_{25} = r_{24} + r_{20}$	66	$r_{66} = r_{62} \cdot r_{62}$
26	$r_{26} = r_{15} \cdot r_{15}$	67	$r_{67} = r_{38} \cdot r_{50}$
27	$r_{27} = r_{25} \cdot r_{26}$	68	$r_{68} = r_{67} \cdot r_{55}$
28	$r_{28} = r_{23} - r_{27}$	69	$r_{69} = r_{65} + r_{66}$
29	$r_{29} = r_{14} \cdot r_{28}$	70	$r_{70} = r_{69} - r_{39}$
30	$r_{30} = r_3 - F$	71	$r_{71} = r_{57} \cdot r_{70}$
31	$r_{31} = r_{30} - Z$	72	$r_{72} = r_{71} + r_{39}$
32	$r_{32} = r_{31} - \alpha$	73	$r_{73} = r_{72} - 1$
33	$r_{33} = r_{32} - 2d$	74	$r_{74} = r_{57} \cdot r_{65}$
34	$r_{34} = r_{33} - Z$	75	$r_{75} = r_{57} \cdot r_{66}$
35	$r_{35} = r_{34} - W_*$	76	$r_{76} = r_{74} \cdot r_{75}$
36	$r_{36} = r_{35} \cdot r_{35}$	77	$r_{77} = 4 \cdot r_{76}$
37	$r_{37} = r_{36} + 1$	78	$r_{78} = r_{73} \cdot r_{73}$
38	$r_{38} = r_{29} \cdot r_{37}$	79	$r_{79} = r_{78} - r_{77}$
39	$r_{39} = y \cdot y$	80	$r_{80} = r_{79} + 1$
40	$r_{40} = h \cdot r_8$	81	$r_{81} = r_{68} \cdot r_{80}$
41	$r_{41} = r_9 - r_{40}$	82	$r_{82} = r_{81} - 1$

The output polynomial is r_{82}.

B Literal shared circuit for the seven-equation system

The same convention applies: the registers are abbreviations unless one deliberately uses the quadratic lift of Section 9. The order of output comparisons in the machine-readable packet differs from the display in (12), but the seven equations are identical.

Table 2: The exact 64-operation circuit for the seven comparisons.

Gate	Polynomial abbreviation	Gate	Polynomial abbreviation
1	$r_1 = \tau \cdot \tau$	33	$r_{33} = h \cdot r_{25}$
2	$r_2 = (B-1) \cdot J$	34	$r_{34} = r_{28} + r_{24}$
3	$r_3 = \eta + \zeta$	35	$r_{35} = r_6 \cdot r_{27}$
4	$r_4 = f \cdot f$	36	$r_{36} = r_{29} + r_3$
5	$r_5 = y \cdot y$	37	$r_{37} = r_{30} \cdot r_{31}$
6	$r_6 = K + w$	38	$r_{38} = 4 \cdot r_{31}$
7	$r_7 = T \cdot f$	39	$r_{39} = r_{31} \cdot r_{31}$
8	$r_8 = r_2 + 1$	40	$r_{40} = r_{30} \cdot r_{30}$
9	$r_9 = r_4 - 1$	41	$r_{41} = r_3 - r_{33}$
10	$r_{10} = t \cdot r_2$	42	$r_{42} = r_{35} + r_{14}$
11	$r_{11} = r_7 - 1$	43	$r_{43} = r_{30} \cdot r_{11}$
12	$r_{12} = r_8 \cdot r_8$	44	$r_{44} = r_{34} \cdot r_4$
13	$r_{13} = w \cdot r_8$	45	$r_{45} = r_{29} \cdot r_{36}$
14	$r_{14} = r_8 - F$	46	$r_{46} = r_{13} + r_{37}$
15	$r_{15} = r_8 \cdot M_F$	47	$r_{47} = r_{38} + 3$
16	$r_{16} = r_{12} \cdot r_8$	48	$r_{48} = r_{41} - r_{34}$
17	$r_{17} = r_{14} - Z$	49	$r_{49} = r_{42} - r_{10}$
18	$r_{18} = r_2 \cdot r_{14}$	50	$r_{50} = i \cdot r_{40}$
19	$r_{19} = r_{12} - 1$	51	$r_{51} = r_{43} - r_{44}$
20	$r_{20} = M_C + r_{15}$	52	$r_{52} = r_1 - r_{45}$
21	$r_{21} = s \cdot r_{16}$	53	$r_{53} = \gamma \cdot r_{47}$
22	$r_{22} = r_{17} - \alpha$	54	$r_{54} = r_{39} + r_{47}$
23	$r_{23} = r_{18} + r_{17}$	55	$r_{55} = r_{51} \cdot r_{51}$
24	$r_{24} = r_{20} \cdot J$	56	$r_{56} = r_{50} \cdot r_{50}$
25	$r_{25} = r_{13} \cdot r_{21}$	57	$r_{57} = r_{46} + r_{53}$
26	$r_{26} = r_3 \cdot r_{21}$	58	$r_{58} = r_{54} \cdot r_{40}$
27	$r_{27} = r_{22} - 2d$	59	$r_{59} = r_{54} \cdot r_9$
28	$r_{28} = r_{23} \cdot r_{19}$	60	$r_{60} = r_{55} - r_5$
29	$r_{29} = r_{25} \cdot r_{26}$	61	$r_{61} = r_{57} \cdot r_{57}$
30	$r_{30} = r_{26} + \eta$	62	$r_{62} = r_{59} \cdot r_{60}$
31	$r_{31} = r_{25} + r_{21}$	63	$r_{63} = r_{61} - r_{58}$
32	$r_{32} = r_{27} - Z$	64	$r_{64} = r_{62} + r_5$

The seven comparisons are $r_{52} = 1,\ r_{63} = 1,\ r_{48} = 1,\ r_{49} = 1,\ r_{64} = 1,\ r_{56} = r_{59},\ r_{32} = W_*$.

References

[1] Vladimir Reshetnikov, *ProveIt*, source snapshot `cdd582bec13f908df3711fa9e05d1ddac368a5db`. https://github.com/VladimirReshetnikov/ProveIt. The full repository paths and SHA-256 values of every included upstream file appear in `vendor/manifest.json`.

[2] ProveIt, *K215 verification README* and *KEndpoint215.lean*, at the pinned snapshot. Pinned K215 verification directory. The source endpoint is `SetTheory.BusyBeaver.ZFC.KEndpoint215.blank_halts_iff_zfc_refutation`.

[3] ProveIt, `FIXED_RAW_UNIVERSAL_75_PROOF.md`, `FIXED_RAW_UNIVERSAL_76_PROOF.md`, and `complete75_half_binomial_compiler.md`, pinned snapshot. The associated layout, no-carry, and actual-index arguments are included with their immediate proof dependencies in `vendor/`.

[4] ProveIt, `FIXED_RAW_UNIVERSAL_78_PROOF.md`, `FIXED_RAW_UNIVERSAL_81_PROOF.md`, and `explore_helical_unary_tableau.py`, pinned snapshot. These define the native window compiler and stationary input markers.

[5] ProveIt, `review_complete85_auxiliary_bezout_math.md`, `review_complete86_ordinary_auxiliary_math.md`, `PELL_RELAXED_AUXILIARY_PROOF.md`, and `complete75_gamma_dominance_elimination102.md`, pinned snapshot.

[6] ProveIt, the complete84 scaled-strong, complete85 reduced-auxiliary, complete86 ordinary-auxiliary, and complete92 norm-unit-guard source packets and accompanying mathematical notes, pinned snapshot. Exact input JSON bytes are included in `vendor/`.

[7] ProveIt, `complete84_signed_quotient_absorption.md`, `complete84_signed_quotient_soundness_scout.md`, and `review_complete84_signed_quotient_soundness.md`, pinned snapshot.

[8] James P. Jones, "Universal Diophantine equation," *Journal of Symbolic Logic* **47** (1982), 549–571. doi:10.2307/2273588.

[9] M. Carl and B. Z. Moroz, "On a Diophantine Representation of the Predicate of Provability," *Journal of Mathematical Sciences* **199** (2014), 36–52. doi:10.1007/s10958-014-1830-2.

[10] A.-M. Legendre, *Essai sur la théorie des nombres*, Paris, 1798. Three-square theorem: a positive integer is a sum of three integer squares if and only if it is not of the form $4^a(8b+7)$. Only this classical theorem is used for the optional unrestricted-integer conversion; the principal positive-variable constructions do not need it.